25901891 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCCOM201 - QUANTITATIVE TECHNIQUES FOR BUSINESS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer any **12** questions

Each question carries **1** marks

1. Which branch of statistics is useful for summarizing sales data of a shop? [A] / [CO1]
2. If the maximum value in a series is 25 and its range is 15,the minimum value of the series is [A] / [CO2]
3. If there are three items, what do you call the cube root of the product of those three items? [U] / [CO1]
4. Which measure of dispersion ignores the signs of deviations? [U] / [CO2]
5. What does kurtosis measure in a data set? [U] / [CO3]
6. What do you call the reciprocal of the arithmetic average of the recirocal of the values of a series? [U] / [CO1]
7. The Coefficient of variation of two series are 58% and 69%. Their Standard Deviations are21.2 and 15.6. What are their Arithmetic Mean [A] / [CO2]
8. How is the transpose of a matrix denoted? [K] / [CO4]

9. What is a stub in a table? [U] / [CO5]

10. Karl Pearsons coefficient of skewness is 0.32.Its SD is 6.5 and mean is 29.6. Find mode. [A] / [CO3]

11. Consider the data: 2,4,6,8 . The Mean Deviation about Mean is [A] / [CO2]

12. In matrices, If A= [-6 -6], B= [-1 -2], find A-B. [A] / [CO4]

13. A square matrix with 1s on the diagonal and 0s elsewhere is called ..... [K] / [CO4]

14. Which type of data were collected through questionnaire? [U] / [CO5]

[1x12 = 12]

Part B

Short Answer/ Problem Type Questions

Answer any **4** questions

Each question carries **7** marks

15. What are the considerations for choice of an average? Comment. [U] / [CO1]

16. Calculate Mean Deviation from arithemetic mean of the given series relating to monthly wages.
Wages (Rs.): 21,23,25,28,30,32,38,39,46,48. [A] / [CO2]

17. Given $\mu_2=9$, $\mu_3=0$, $\mu_4=200$, analyze skewness and kurtosis. [A] / [CO3]

18. ABC Trading Company has two product lines: Product A and Product B. The company operates in two markets: Urban and Rural. The profit margins (in percentage) and market share data are organized as follows: The profit margin matrix P is given by: Urban market: Product A has 15% profit margin, Product B has 20% profit margin Rural market: Product A has 12% profit margin, Product B has 18% profit margin. Set up the profit margin matrix P as a 2×2 matrix where rows represent markets (Urban, Rural) and columns represent products (A, B). Also find the adjoint of matrix P. [A] / [CO4]

19. Distinguish between Census Method and Sampling Method. [U] / [CO5]

20. The analysis of the results of a budget survey of 150 families gave an average monthly expenditure of Rs. 120 on food items. After the analysis was completed, it was noted that the figure recorded for one household was wrongly taken as Rs. 15 instead of Rs. 105. Determine the correct value of the average expenditure. [A] / [CO1]

[7x4 = 28]

Part C

Essay/ Problem Type Questions

Answer any **2** questions

Each question carries **15** marks

21. Calculate Median and Mode from the following data: [A] / [CO1]

Class	0-7	7-14	14-21	21-28	28-35	35-42
Frequency	7	11	24	19	12	9

22. The distribution of marks are given below. Compute the Mean, Standard Deviation and Coefficient of Variation. [A] / [CO2]

Marks	**No. of Students**
0–20	5
20–40	15
40–60	20
60–80	6
80–100	4
Total	**50**

23. XYZ Retail Company sells three product categories (Electronics, Clothing, Furniture) across 4 different stores. The company wants to calculate the total daily sales commission for each store. Sales Matrix S (units sold per store): The units sold by each store are as follows (Electronics, Clothing & Furniture respectively): Store 1: 50, 120, 80 Store 2: 40, 150, 90 Store 3: 60, 100, 110 Store 4: 45,130, 95 Commission Matrix C (commission per unit in rupees): The commission for each product category is as follows: (Electronics, Clothing & Furniture respectively): Store 1: 100,50, 200 Store 2: 120, 60, 250 Store 3: 110, 55, 80 Store 4: 90, 65, 80 [A] / [CO4]

    a) Prepare the Sales Matrix S and Commission Matrix C.

    b) Calculate the daily sales commission for each store.

    c) Which store has the highest total sales commission?

    d) Which store has the lowest total sales commission?

[15x2 = 30]